

MATH 6241 – Stochastic Processes

❖ Section 4. Classification of States and Chains

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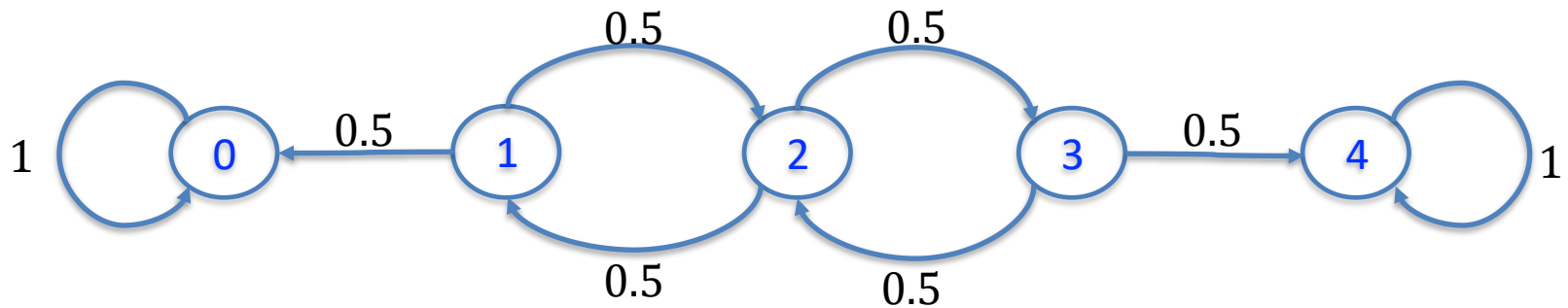
Structure of Chains

- Communication
- classes and absorbing states
- Transient vs recurrent states
- Renewal Equation
- Irreducible chains

Example: Drunkard's walk

The state space is $\Omega = \{0, 1, 2, 3, 4\}$, and X_n is the drunkard's position after n steps.

At each step he goes left or right with probability $\frac{1}{2}$ until he reaches an endpoint 0 or 4, where he stays forever.



A finite state Markov chain can be usefully represented by a directed graph where the vertices are the states, and edges are the allowed one-step transitions.

The transition mechanism depends only on the current state, which means that this is a Markov chain.

The transition matrix is

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Recall:

- A state j is called **accessible from state i** , if it is possible to move from i to j in n steps, i.e., $p_{ij}(n) = (P^n)_{ij} > 0$ for some $n \geq 0$.
- A state i is called **absorbing**, if $p_{ii} = 1$. That is, once the system hits state, it stays there forever.

- The end states $\{0,4\}$ of the drunkard's walk are *absorbing* states.

Question: Are all states accessible from all other states?

- State $\{1,2,3\}$ are *not accessible* from state 0 or 4.

$$\begin{aligned}P(X_3 = 1 | X_0 = 2) &= P(2 \rightarrow * \rightarrow * \rightarrow 1) \\&= P(2 \rightarrow 1 \rightarrow 2 \rightarrow 1) + P(2 \rightarrow 3 \rightarrow 2 \rightarrow 1) \\&= \frac{1}{8} + \frac{1}{8} = \frac{1}{4}\end{aligned}$$

$$P(X_3 = 1 | X_0 = 2) = (P^3)_{21}$$

□ Communication

State $j \in \Omega$ is called **accessible** from state i , written $i \rightarrow j$, if

$$\mathbb{P}(X_n = j | X_0 = i) =: p_{ij}^{(n)} = (P^n)_{ij} > 0 \text{ for some } n \geq 0.$$

States $i, j \in \Omega$ is called **communicate**, written $i \leftrightarrow j$, if

$$i \rightarrow j \text{ and } j \rightarrow i$$

Proposition (Communication is an Equivalence Relation)

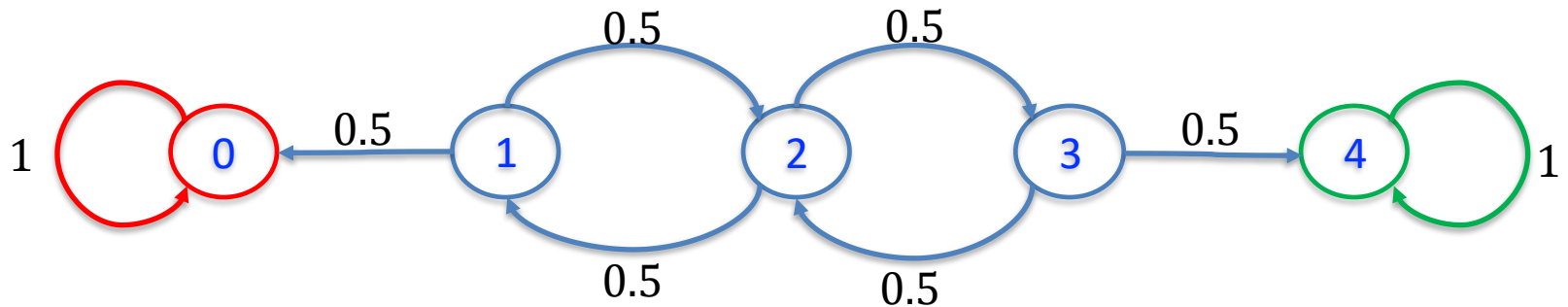
- **Reflexive:** $p_{ii}^{(0)} = 1 \Rightarrow i \leftrightarrow i$
- **Symmetric:** by definition
- **Transitive:** if $i \rightarrow j$ and $j \rightarrow k$, then $i \rightarrow k$

$$p_{ik}^{(m+n)} = \sum_t p_{it}^{(m)} p_{tk}^{(n)} \geq p_{ij}^{(m)} p_{jk}^{(n)} > 0$$

Communicating classes

The state space decomposes **uniquely** into **communicating classes**.

Example: Drunkard's walk



- Absorbing states are the states for which $p_{ii} = 1$. So once the chain enters an absorbing state it can never leave. Absorbing states form **singleton** communicating classes

□ First visit and eventually visit probability

Consider the *Discrete-time Markov chain*:

$$\begin{array}{c} i \\ \parallel \\ X_0 \end{array} \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow X_{n-1} \rightarrow \begin{array}{c} j \\ \parallel \\ X_n \end{array} \rightarrow X_{n+1}, \dots$$

Let $f_{ij}(n)$ to be the **probability** that starting in state i the chain **first visits** state j after n steps.

$$f_{ij}(n) := \mathbb{P}(X_1 \neq j, X_2 \neq j, \dots, X_{n-1} \neq j, X_n = j \mid X_0 = i)$$

Let f_{ij} to be **probability** that the chain **eventually visits** state j starting in state i .

$$f_{ij} := \sum_{n=1}^{\infty} f_{ij}(n) = P(X_n = j \text{ for some } n \geq 1 \mid X_0 = i)$$

□ First Return Times and Return Probabilities

Given $X_0 = i$, the **First Return Time**:

$$T_i := \inf_n \{n \geq 1 \mid X_n = i\}$$

First return probability: $f_{ii}^{(n)} := \mathbb{P}(X_1 \neq i, X_2 \neq i, \dots, X_{n-1} \neq i, X_n = i \mid X_0 = i)$
 $= \mathbb{P}(T_i = n \mid X_0 = i)$

Total Return Probability: $f_{ii} = P(\text{return to } i \mid X_0 = i)$

$$= P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i)$$

$$= \sum_{n=1}^{\infty} f_{ii}^{(n)}$$

$$= \mathbb{P}(T_i < \infty)$$

Two types of states: Recurrence and Transience (Will we come back?)

Definition: The state i is **recurrent** (persistent) if $f_{ii} := \mathbb{P}(T_i < \infty) = 1$.

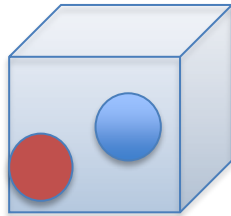
The state i is **transient** if $f_{ii} := \mathbb{P}(T_i < \infty) < 1$.

- The state i is **recurrent** if the chain must eventually return to state i , given that it started at state i .

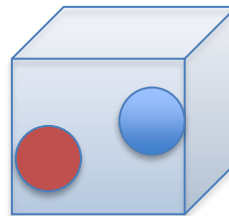
Since the clock starts over at each return to state i , we conclude that if the state i is persistent then the chain must return to i infinitely often.

- If state i is **transient**, then the chain will return to state i only finitely often. Eventually it will never return to the state. (Temporary in i .)
- Absorbing state is a special recurrent state.

Continue Example (Balls and Boxes Markov Chain) .



Box1



Box2

Let X_n denote the number of red balls in the first box after n steps.

p_{ij} = transition probability $i \rightarrow j$, where $i, j \in \Omega = \{0, 1, 2\}$ are possible states.

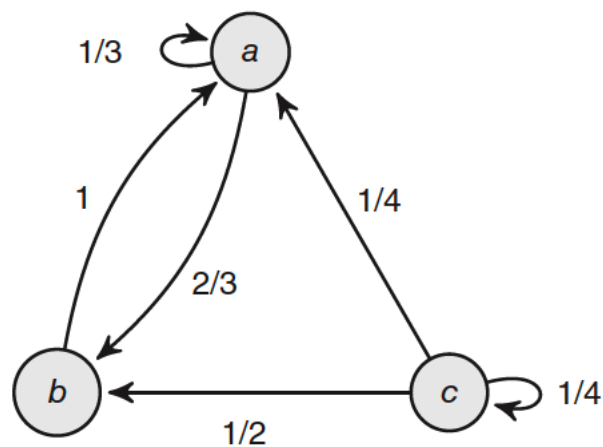
Transition Matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$



Question: Is State 1 *recurrent*?

Example:



$$P = \begin{bmatrix} 1 & 2 & \\ \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

Renewal Equation for Markov Chains

Probability that the chain is at i at time n , allowing multiple earlier returns.

$$p_{ii}^{(n)} := \mathbb{P}(X_n = i \mid X_0 = i)$$

Decomposition of as disjoint Events, by first return

$$\{X_n = i\} = \bigcup_{k=1}^n \{T_i = k, X_n = i\}$$

Conditioning on the First Return Time, by Strong Markov Property

$$\begin{aligned} \mathbb{P}(X_n = i \mid T_i = k, X_0 = i) &= \mathbb{P}(X_n = i \mid X_k = i) \\ &= \mathbb{P}(X_{n-k} = i \mid X_0 = i) \\ &= p_{ii}^{(n-k)} \end{aligned}$$

The Renewal Equation

$$\overset{i}{\parallel} X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow X_{n-1} \rightarrow \overset{i}{\parallel} X_n \rightarrow X_{k+1}, \dots$$

$$p_{ii}^{(n)} := \mathbb{P}(X_n = i | X_0 = i)$$

$$= \sum_{k=1}^n \mathbb{P}(T_i = k, X_n = i | X_0 = i)$$

$$= \sum_{k=1}^n \mathbb{P}(X_n = i | T_i = k, X_0 = i) \mathbb{P}(T_i = k | X_0 = i)$$

$$= \sum_{k=1}^n p_{ii}^{(n-k)} f_{ii}^{(n)}$$

Initial condition: $p_{ii}^{(0)} = \mathbb{P}(X_0 = i | X_0 = i) = 1$

□ Generating-Function Form

Define power series:

$$H(z) = \sum_{n=0}^{\infty} p_{ii}^{(n)} z^n$$

$$F(z) = \sum_{n=0}^{\infty} f_{ii}^{(n)} z^n$$

Multiply the renewal equation by z^n and sum over $n \geq 1$:

$$H(z) - 1 = F(z)H(z)$$

So, **Generating-Function Form of the renewal equation**

$$H(z) = \frac{1}{1 - F(z)}$$

In particular, the total return probability $f_{ii} := \sum_{n=1}^{\infty} f_{ii}^{(n)} = F(1)$

□ Expected number of visits

Given $X_0 = i$, the total number of visits to state i is defined by

$$N_i := \sum_{n=0}^{\infty} \mathbb{I}(X_n = i) \quad \text{where } \mathbb{I} \text{ is the indicator function.}$$

The **expected number of visits to i** is

$$E(N_i) = \sum_{n=0}^{\infty} E(\mathbb{I}(X_n = i)) = \sum_{n=0}^{\infty} \mathbb{P}(X_n = i | X_0 = i) = \sum_{n=0}^{\infty} p_{ii}^{(n)} = H(1)$$

From renewal equation, the expected number of visits to i is

$$E(N_i) = H(1) = \frac{1}{1 - F(1)} = \frac{1}{1 - f_{ii}} = \frac{1}{1 - \sum_{n=1}^{\infty} f_{ii}^{(n)}}$$

This gives a useful way to decide if a state is recurrent or transient.

□ Deciding about recurrent (by expected number of visits)

Theorem: The state i is **recurrent** if and only if expected number of visits to i

$$E(N_i) = \sum_{n=1}^{\infty} P(X_n = i | X_0 = i) = \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty$$

Equivalently, using power of transition matrix, the state i is recurrent if and only if

$$\sum_{n=1}^{\infty} (P^n)_{ii} = \infty$$

Equivalently, the state i is **recurrent** if and only if expected number of visits to i is **infinite**.

If state j is **transient**, then the expected number of visits to j is **finite**:

$$H(1) = \frac{1}{1 - F(1)} = \frac{1}{1 - f_{jj}}$$

If state j is transient,

$$\sum_{n=1}^{\infty} (P^n)_{jj} \quad \text{implies that} \quad \lim_{n \rightarrow \infty} (P^n)_{jj} = 0$$

Moreover, if j is accessible from i , then

$$\sum_{n=1}^{\infty} (P^n)_{ij} \quad \text{implies that} \quad \lim_{n \rightarrow \infty} (P^n)_{ij} = 0$$

□ Classification of Markov chains

Theorem: Suppose i, j communicate then they are either both recurrent or both transient. That is,

$$i \text{ is recurrent} \Leftrightarrow j \text{ is recurrent}$$

$$i \text{ is transient} \Leftrightarrow j \text{ is transient}$$

Recurrence/transience depends only on the communicating class.

Irreducible Markov Chain

If all states communicate, we say that the Markov chain is **irreducible**.

Corollary:

- Every finite Markov chain has at least one recurrent state.
- For a **finite** irreducible Markov chain, all states are recurrent.

Simple Random walk

Example: Simple Random walk on integers $\mathbb{Z}$ starting at 0.

The Chain is irreducible, however, all states are transient if $p \neq \frac{1}{2}$.

Example: Simple Random walk on $\mathbb{Z}^2$ starting at 0.

Example: Simple Random walk on $\mathbb{Z}^3$ starting at 0.

‘A Drunk man will find his way home, but a drunk bird may get lost forever. ‘

Proof of the theorem:

Since i, j communicate, there are integers n, m such that

$$i \xrightarrow{n} j \xrightarrow{m} i \qquad h = p_{ij}^{(n)} p_{ji}^{(m)} > 0$$

Hence, for any r ,

$$p_{ii}^{(n+m+r)} = \mathbb{P}(X_{n+m+r} = i | X_0 = i) \geq p_{ij}^{(n)} p_{jj}^{(r)} p_{ji}^{(m)} = h p_{jj}^{(r)}$$

$$\begin{array}{ccccccc} & n & & r & & m & \\ i & \xrightarrow{\quad} & j & \xrightarrow{\quad} & j & \xrightarrow{\quad} & i \\ X_0 & & X_n & & X_{n+r} & & X_{n+r+m} \end{array}$$

Sum over r ,

$$\sum_k p_{ii}^{(k)} \geq \sum_r p_{jj}^{(n+m+r)} \geq h \sum_r p_{jj}^{(r)}$$

Therefore either both sums are finite or both are infinite, hence either both states are transient or both are persistent.

Example (Balls and Boxes Markov Chain) .

Let X_n denote the number of red balls in the first box after n steps.

Transition Matrix

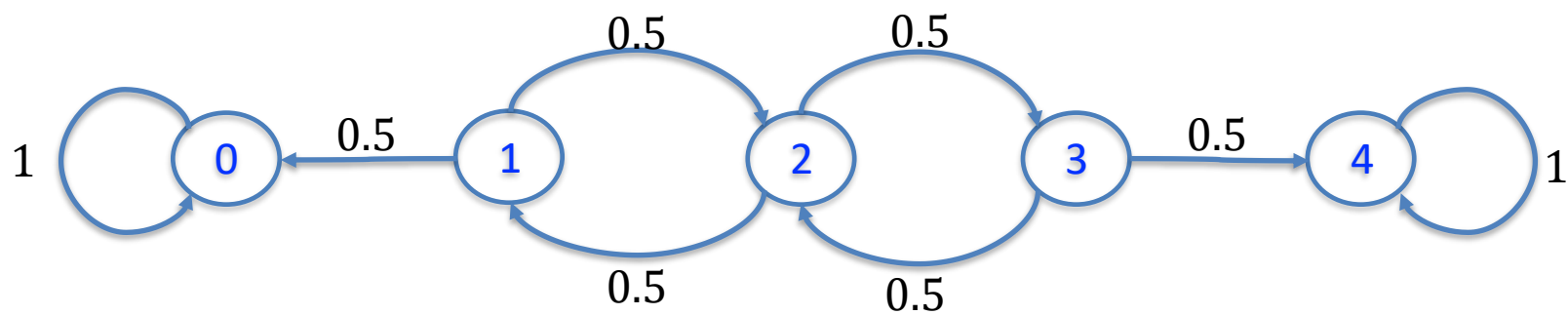
$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$



States 0, 1 communicate?

States 0, 2 communicate?

Example: Drunkard's walk



Decomposition of the state space.

Let (X_n) be a discrete-time Markov chain on a countable state space Ω with transition probabilities p_{ij} .

A communicating class of states C in Ω is called **closed**, if

$$p_{ij} = 0 \text{ whenever } i \in C \text{ and } j \notin C.$$

The state space Ω can be **partitioned uniquely** as

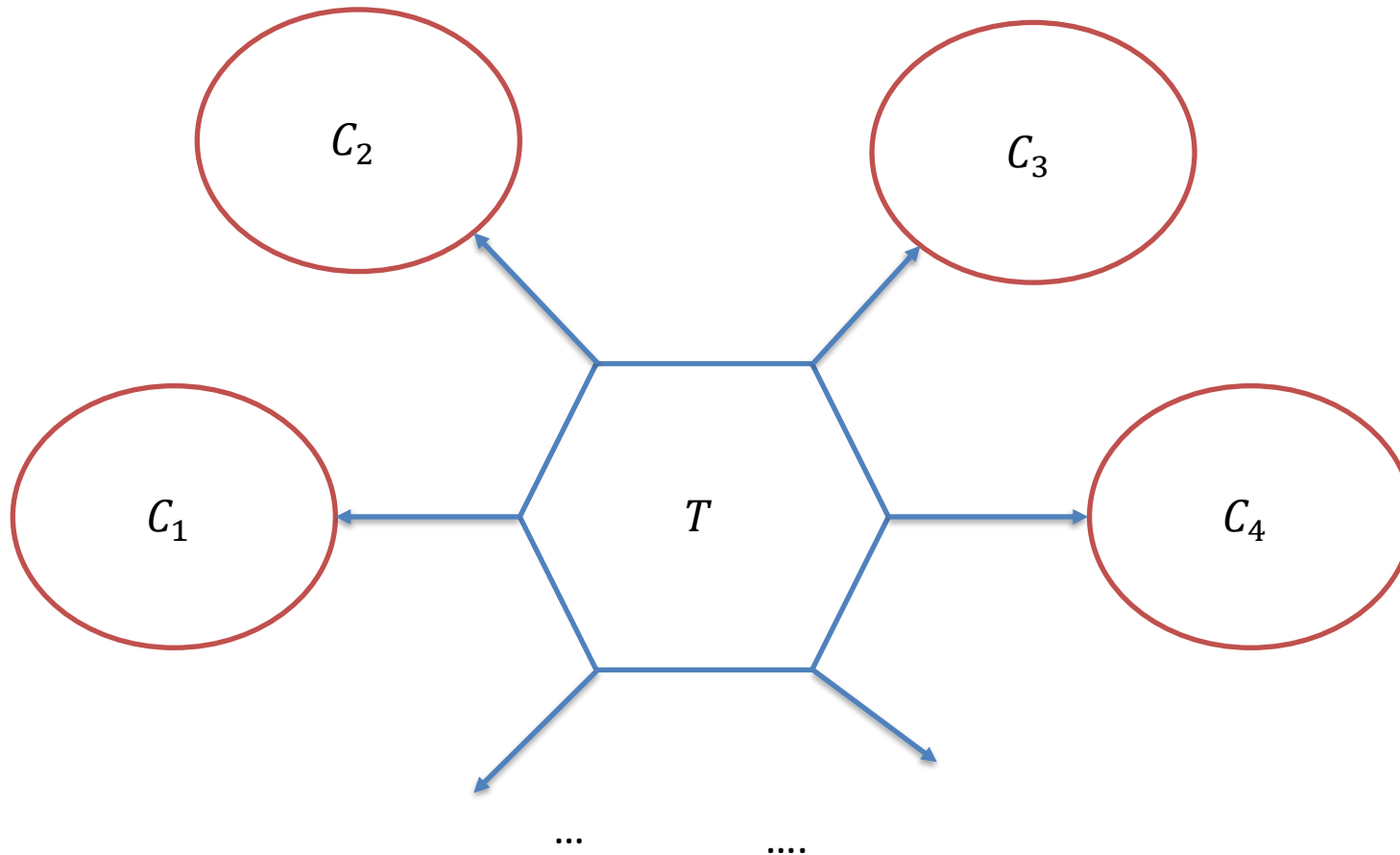
$$\Omega = T \cup C_1 \cup C_2 \cup \dots$$

where

- T is the set of all **transient states**;
- each C_k is a **closed, irreducible communicating class**; (recurrent)
- the sets $T, C_1, C_2, \dots$ are **disjoint**

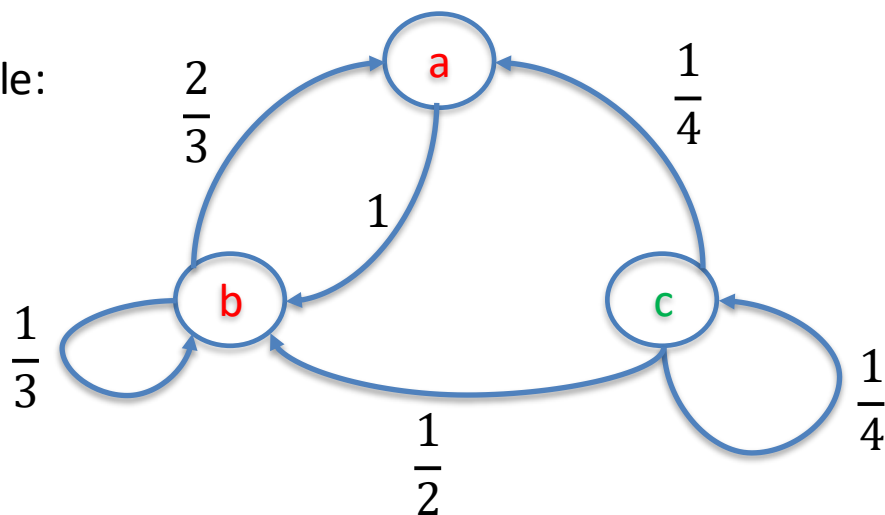
Relabel (permute) the states $\Omega = T \cup C_1 \cup C_2 \cup \dots \cup C_r$, the transition matrix is of the block upper-triangular form:

$$P = \begin{pmatrix} Q & R_1 & R_2 & \cdots & R_r \\ 0 & P_1 & 0 & \cdots & 0 \\ 0 & 0 & P_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & P_r \end{pmatrix}$$



- If the chain starts with $X_0 \in C_i$ then it stays in C_i forever.
- If it starts with $X_0 \in T$ then eventually it enters one of the classes C_i and stays there forever.
- So the irreducible classes determine the long-time behavior.

Example:



$T = \{c\}$ transient

$C_1 = \{a, b\}$ closed, irreducible

Order states as $\{c, a, b\}$. Then

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} Q & R \\ 0 & P_1 \end{pmatrix}$$

Positive and Null Recurrence (How fast do we come back?)

Mean return time: For a **recurrent** state i , define

Recall that given $X_0 = i$, the First Return Time:

$$T_i := \inf_n \{n \geq 1 \mid X_n = i\}$$

So, Expected number of steps to **return to state** i , starting from i is

$$r_i := E[T_i]$$

Definitions

- The state i is called **positive recurrent** if $r_i < \infty$
- The state i is called **null recurrent** if $r_i = \infty$

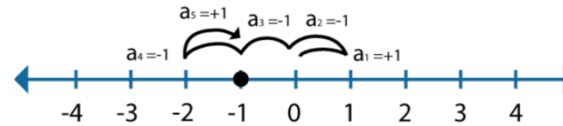
Classification of States:

$$\text{States} = \begin{cases} \text{Transient} \\ \text{Recurrent} \end{cases} = \begin{cases} \text{Positive recurrent} \\ \text{Null recurrent} \end{cases}$$

There is **no overlap** and **no missing case**.

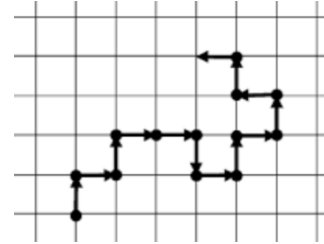
Canonical Examples

(A) Simple Random Walk on $\mathbb{Z}$



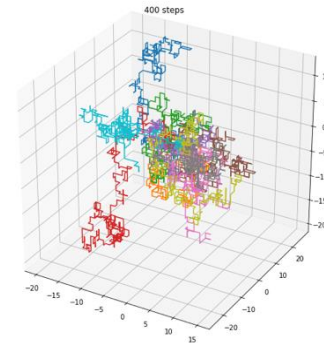
Null recurrent

(B) Simple Random Walk on $\mathbb{Z}^2$



Null recurrent

(C) Simple Random Walk on $\mathbb{Z}^3$



Not recurrent

(D) Finite State Irreducible Markov Chain

All states are positive recurrent

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross